

Ayushmaan Puri

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EDUCATION

University of California, San Diego

B.S. CSE: Computer Engineering

September 2024 - September 2026

GPA 3.40/4.0

Orange Coast College

A.S. for Transfer: Computer Science, Mathematics, and Physics – Graduated with Honors

August 2022 - May 2024

GPA 3.87/4.0

RESEARCH EXPERIENCE

Undergraduate Researcher | **Junkyard Computing**

Kastner Research Group – UC San Diego

March 2026 – September 2026

La Jolla, CA

Batteryless Operation and Thermal Testing

June 2026 – September 2026

- Enabled batteryless operation of stripped-down Pixel Fold nodes by wiring regulated DC power to the mainboard and spoofing the battery ID and NTC thermistor signals, found by reverse-engineering the dual Samsung PMIC (s2mpg12/s2mpg13) boot sequence in the felix/gs201 kernel tree; this frees the USB-C port for networking and removes thousands of lithium cells from the rack.
- Ran per-cluster **stress-ng** workloads while logging thermal zones (`/sys/class/thermal/`), on-device ODPM power rails, and wall-side power from the bench PSU over its USB interface, to physically locate the Tensor G2 CPU/GPU clusters on the mainboard and map hotspots for cooling design and throttling prevention.
- Coordinating networking and deployment logistics to integrate the phone racks into UCSD's DSMLP data center for course autograding, working across the thermal/rack, teardown, and kernel teams and the data center staff.

Autograder Cluster

March 2026 – June 2026

- Investigated repurposing discarded smartphones as a dedicated autograding cluster to replace oversubscribed GPU pods shared across a class of 330 students.
- Deployed a 40-node k3s Kubernetes cluster on ARM64 Google Pixel Fold devices running Debian over USB Ethernet; configured each node as an isolated Linux execution environment for containerized grading jobs.
- Modeled real submission timing and runtime distributions from 8 programming assignments using Gaussian Mixture Models with BIC-selected component counts.
- Built a synthetic workload generator sampling from fitted GMMs to stress test cluster throughput and determine the minimum phone count needed to serve a class without queuing delays.

Undergraduate Researcher | **UnifiedSplitting**

PatLab & UC Scholars Research Program – UC San Diego

June 2025 – February 2026

La Jolla, CA

- Built heterogeneous CPU-GPU pipelines for layer-split CNN inference on NVIDIA Jetson unified-memory hardware.
- Profiled and characterized per-layer CPU-GPU speedup ratios across full network architectures; identified a 10x performance gap between convolutional layers (300–500x) and fully connected layers (30–40x) to determine optimal split points.
- Validated the approach across five architectures (ResNet-18/50, MobileNetV2, EfficientNet-B2, ViT-Tiny) and six simulated edge hardware profiles, achieving 8–24% throughput gains over GPU-only baselines.

Summer Research Intern | **Active Thermography for Defect Detection**

SeNSE – Indian Institute of Technology, Delhi

June 2024 – Sep. 2024

New Delhi, India

- Developed a per-pixel signal processing pipeline in Python (OpenCV, NumPy, SciPy) to detect subsurface defects in Carbon and Glass Fibre-Reinforced Polymer (CFRP/GFRP) composites using active thermography.
- Implemented linear detrending, full temporal cross-correlation against a sound reference pixel, main lobe extraction, and FFT phase computation across three experimental datasets (LFM, Active Pulsed, and Flux excitation).
- Demonstrated that the FFT phase of the correlation main lobe produces qualitatively distinct signatures for defective vs. healthy material regions, establishing a new discriminating observable for non-destructive evaluation. Recorded results and findings in a [research paper](#).

Summer Research Intern | **Lunar Rover Modeling**

NASA California Space Grant Consortium

June 2023 – Aug. 2023

Costa Mesa, CA

- Designed and built a model lunar lava tube exploration rover using a rocker-bogie chassis, solar-powered with lithium-polymer battery storage.
- Developed Arduino firmware integrating UV, Hall effect magnetic, and gravimeter sensors to detect water-ice signatures, iron-rich rock, and subsurface density anomalies.
- Implemented on-board SD card data logging for sensor telemetry and an object avoidance drive system for autonomous navigation over loose regolith.

PUBLICATIONS AND TALKS

UnifiedSplitting: CPU–GPU Split Inference on Edge Devices <i>Oral presentations</i>	2025
- Southern California Conference for Undergraduate Research (SCCUR), CSU Channel Islands, Nov. 2025, with Fahad Alkhazam and Parth Mehta. [Slides] [Abstract] - UC San Diego Summer Research Conference (SRC), Aug. 2025. [Abstract]	

AWARDS AND HONORS

CRA Outstanding Undergraduate Researcher Honorable Mention <i>Computing Research Association</i>	Jan. 2026
UC Scholar (\$5000) <i>UC San Diego</i>	May 2025
George Ciarlo Memorial Scholarship (\$1000) <i>Orange Coast College</i>	Apr. 2024

TEACHING, MENTORSHIP, AND SERVICE

Research Mentor – Engineers for Exploration (E4E)	June 2026 – Aug. 2026
<i>Matthew Henson Fellowship – UC San Diego</i>	<i>La Jolla, CA</i>
- Mentored visiting undergraduate researchers from Howard University on the Junkyard smartphone cluster project. - Guided students through SoC stress-testing on repurposed Pixel devices and custom server rack design for logic boards.	

Conference Moderator	August 2026
<i>UC San Diego Summer Research Conference 2026</i>	<i>La Jolla, CA</i>
Moderated two sessions with 4 presenters each. Computer Science for the Public Good and Research in Electrical, Computer, and Mechanical Engineering	

STEM Peer Mentor	Aug. 2023 – May 2024
<i>OCC STEM Academy / MESA Program</i>	<i>Costa Mesa, CA</i>
- Mentored and tutored 17 first-generation STEM students, providing weekly one-on-one guidance, email check-ins, and long-term academic planning support. - Organized and led workshops on career development, internships, and transfer preparation, increasing student awareness of opportunities beyond community college. - Co-managed daily operations of the STEM/MESA Center with 6 peer mentors, coordinating tutoring schedules, student outreach, and resource distribution.	

Club President	Aug. 2023 – May 2024
<i>OCC STEM Club</i>	<i>Costa Mesa, CA</i>
- Hosted weekly workshops and projects for 20+ members across all STEM disciplines, fostering an inclusive community for students to explore diverse scientific interests. - Organized volunteering missions to foster interest in STEM fields among children. Led a group of 10 volunteers to teach young students about physics and biology at OCC’s annual science night.	

Volunteer Teaching Assistant	July 2023 – Feb. 2024
<i>Microsoft TEALS Program</i>	<i>Houston, TX (Remote)</i>
- Co-taught CS Discoveries curriculum twice weekly to a class of 26 middle school students, integrating lessons on programming logic and problem-solving. - Designed a sustainable CS education framework for Santa Fe Junior High, enabling the program to continue beyond the volunteer period.	

Honors Mathematics Tutor	Jan. 2023 – June 2023
<i>OCC Student Success Center</i>	<i>Costa Mesa, CA</i>
Tutored 10–12 students daily in calculus (single and multi-variable), precalculus, trigonometry, and college algebra	

PROJECTS

Tiled Matrix Multiplication Visualizer <i>HTML, JS, Parallel Programming</i>	Feb. 2026
Built an interactive visualizer, deployed at aypuri.com/tiledmatmultvis, to demonstrate tiled matrix multiplication, data reuse, and memory access patterns.	

ESP32 Bluetooth Speaker <i>Microcontrollers, Circuit Design, C++</i>	Apr. 2025
Built a Bluetooth speaker with a display showing title, artist, and album, powered by an ESP32 microcontroller.	

DSA Visualizer <i>TypeScript, Web Development</i>	Apr. 2024
Co-created a web application, deployed at animated-octo-spork.vercel.app, with a TypeScript back-end to help students visualize core data structures (linked lists, trees, graphs, stacks, queues).	